

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks: 30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I VISHAL.P (192512340) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P. Vishal

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MP

C03 - AT3 Resolution Algorithm Implementation

1. Premium Subscription Access:-

1. Let

(i) P = customer pays the subscription fee

(ii) A = customer gets premium access.

(iii) P_a = Aithe paid the fee

(iv) A_a = Aithe gets access.

(i) Statement 1:-

If a customer pays the sub fee

$$P \rightarrow A$$

Statement 2:- Aithe paid the subscription fee

P_a

Goal: A_a

ii) Implication rule: $P \rightarrow A \equiv \neg P \vee A$

$\therefore$ The CNF clauses are:

Clause 1: $\neg P_a \vee A_a$

Clause 2: P_a

Negate: $\neg A_a$.

(ii) reserve clause 1 and clause 2:

$$(\neg P_a \vee A_a), P_a$$

Resolving P_a and $\neg P_a$
 A_a .

New clause:

clause A:

$$A_a$$

Now reserve clause A with the neg goal

$$A_a, \neg A_a$$

iv) Aritho gets premium access.

2. Smart device connectivity :-

Let:- w : Smart TV connected to wifi

I : Can access the internet

(i) Propositional logic.

(i) If the smart TV is connected to wifi then it can access the internet

(ii) connected to wifi: $w \rightarrow I$
 w

Goal: I .

ii) CNF and Negated Goal:- $w \rightarrow I \equiv \neg w \vee I$

clauses 1. $\neg w \vee I$

2. w

3. negated goal: $\neg I$

Review done. A and L.

$(\neg w \vee I), w$

T

T, T T

□

Conclusion: Smart TV can access the internet.

Machine Auto shutdown.

Let

O = machine overheats

S = Machine shutdown

$O \rightarrow S$

O

Goal: S

$O \rightarrow S \equiv \neg O \vee S$

$(\neg O \vee S), O$

$\therefore S$

S, TS

□

$\therefore$ The machine shutdown automatically

Account verification :-

Let V = user uploads a valid ID

A = Account is verified

$V \rightarrow A$

V

Goal: A.

$$V \rightarrow A \equiv \neg V \vee A$$

$$(\neg V \vee A), V$$

$$\begin{array}{c} A \\ A, \neg A \end{array}$$

The proved is verified

5. Exam Hall Ticket Eligibility :-

Let
 C = Student completes all assignments
 E = Student is eligible for exam.
 H = Student receives Hall ticket.

$$C \rightarrow E, E \rightarrow H, C$$

$$\neg C \vee E$$

$$\neg E \vee H$$

$$\begin{array}{c} C \\ \neg H \end{array}$$

Resolution steps

$$(\neg C \vee E), C \Rightarrow E$$

$$(\neg E \vee H), E \Rightarrow H$$

$$H, \neg H \Rightarrow \boxed{\square}$$

Conclusion: The student receives a Hall ticket.